

Esperimento 1 - timing policies

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Experiment 1

The objective of this experiment is to evaluate whether an adversary can force an ATS to emit a specific trading signal at a chosen timestep. Instead of measuring attack effectiveness only through final portfolio performance, the experiment directly measures whether the attack achieves its intended operational goal: forcing a **BUY** or **SELL** signal.

Hypothesis

! The main hypothesis is that targeted attacks are more successful when they are launched at timesteps where the ATS decision is already fragile.

In particular, the experiment tests whether attack success depends on the timing policy used to select attack days. The expectation is that attacks near the strategy decision boundary should require smaller perturbations and should have a higher success rate than attacks launched at uniformly selected timesteps or during generic high-volatility periods.

A secondary hypothesis is that market-based indicators, such as price volatility or trend strength, may not be sufficient to identify vulnerable attack points unless they are directly related to the ATS decision rule.

ATS Setups

Three ATS setups were tested:

1. **LSTM_ALL_MA**

This setup uses all tickers in the stock universe and applies a **Moving Average** trading strategy. The attacked asset is **GOOGL**.

2. **LSTM_ALL_ROC**

This setup uses all tickers in the stock universe and applies a **Rate of Change** trading strategy. The attacked asset is **GOOGL**.

3. **LSTM_ALL_BB**

This setup uses all tickers in the stock universe and applies a **Bollinger Bands** trading strategy. The attacked asset is **GOOGL**.

Attack Objective

For each selected attack day, two targeted objectives are evaluated:

- force_buy** : force the ATS to emit a **BUY** signal;
- force_sell** : force the ATS to emit a **SELL** signal.

The attack modifies the normalized **close** feature in the input sequence of the attacked ticker. A grid of perturbation values is tested (from -0.35 to 0.35), and the experiment records the smallest perturbation that causes the strategy to output the target signal.

An attack is considered successful when the attacked signal matches the target signal.

Metrics

The experiment reports two versions of Attack Success Rate.

The first is the **raw Attack Success Rate**:

```
ASR = successful attacks / total attacks
```

However, this metric can be misleading because in some cases the baseline signal is already equal to the target signal before the attack.

For this reason, the most important metric is the **non-trivial Attack Success Rate**:

```
Non-trivial ASR =  
successful attacks where baseline was not already target /  
cases where baseline was not already target
```

This metric measures whether the attack actually changed the ATS decision.

The experiment also records:

- `min_abs_delta_norm`: the minimum normalized perturbation required for success;
- `prediction_shift`: the induced change in the model prediction;
- `delta_final_cr`: the change in final cumulative return.

Timing Policies

Five timing policies were evaluated:

1. `uniform`

Attack days are selected uniformly across the test period. This serves as a baseline.

2. `high_volatility`

Attack days are selected where the rolling volatility of real prices is highest. This represents a market-aware attacker using only observable market data.

3. `high_pred_volatility`

Attack days are selected where the rolling volatility of model predictions is highest. This assumes the attacker has access to the ATS predictions or to a surrogate model.

4. `strong_trend`

Attack days are selected where the absolute price trend is strongest.

5. `near_strategy_boundary`

Attack days are selected where the strategy is closest to its decision threshold. This is a strategy-aware, white-box timing policy.

Results

! The results show that vulnerability of an adversarial attack strongly depends on the trading strategy adopted by the attacked ATS.

LSTM_ALL_MA

For the `LSTM_ALL_MA` setup:

timing_policy	nontrivial_trials	nontrivial_successes	mean_min_abs_delta_norm	nontrivial_attack_success_rate
<code>near_strategy_boundary</code>	14	14	0.013750	1.000000
<code>high_volatility</code>	12	6	0.032778	0.500000
<code>high_pred_volatility</code>	12	4	0.028125	0.333333
<code>uniform</code>	12	3	0.036000	0.250000

timing_policy	nontrivial_trials	nontrivial_successes	mean_min_abs_delta_norm	nontrivial_attack_success_rate
strong_trend	12	2	0.010000	0.166667

This strongly supports the hypothesis that attacks are more effective near the ATS decision boundary. Generic market indicators such as volatility and trend did not identify vulnerable timesteps in this experiment.

The perturbation required for these successful attacks was often minimal, with an average normalized perturbation of just **0.01375** for the **near_strategy_boundary** policy. This data, combined with a **100% non-trivial ASR**, confirms that when the ATS is already near a decision boundary, an extremely small input manipulation is sufficient to force the desired signal.

In contrast, other timing policies required significantly greater effort or achieved far more limited success.

LSTM_ALL_ROC

timing_policy	nontrivial_trials	nontrivial_successes	mean_min_abs_delta_norm	nontrivial_attack_success_rate
high_pred_volatility	16	0	0.0	0.0
high_volatility	15	0	0.0	0.0
near_strategy_boundary	13	0	0.0	0.0
strong_trend	13	0	0.0	0.0
uniform	13	0	0.0	0.0

The results demonstrate that this ATS setup exhibits significant robustness against the current targeted attack configurations. Across all evaluated timing policies, the **nontrivial attack success rate remained at 0.0%**, indicating that the adversary failed to force a signal change in any instance where the baseline signal did not already match the target objective.

The same experiment was tested again with `max_abs_delta=0.5` instead of `0.35`, but the results didn't change. (`max_abs_delta` defines the maximum magnitude of the perturbation, or in other words, "how large" the EP can be).

LSTM_ALL_BB

timing_policy	nontrivial_trials	nontrivial_successes	mean_min_abs_delta_norm	nontrivial_attack_success_rate
high_pred_volatility	14	0	0.0	0.0
high_volatility	13	0	0.0	0.0
near_strategy_boundary	21	0	0.0	0.0
strong_trend	18	0	0.0	0.0
uniform	19	0	0.0	0.0

The experimental results for the **LSTM_AII_BB** setup confirm that the Bollinger Bands (BB) trading strategy is highly robust against the tested targeted attack configurations. Similar to the findings in the ROC experiment, the **nontrivial attack success rate is 0.0%** across all timing policies, indicating that the adversary was unable to force a change in the trading signal in any case where the target was not already met by the baseline.

Interpretation

The experiment suggests that targeted attacks are not simply more successful during volatile or strongly trending market periods. Instead, success appears to depend on the internal decision state of the ATS and the specific logic of the underlying trading strategy.

Strategy-Dependent Vulnerability: For the **Moving Average** strategy, the most vulnerable timesteps are those where the strategy is close to switching signals (near the decision boundary). In these instances, a minimal perturbation in the input is sufficient to move the system across the decision threshold.

Structural Robustness: Conversely, strategies with wider temporal windows and volatility-based parameters, such as **Rate of Change** and **Bollinger Bands**, exhibit significant intrinsic robustness. For these setups, the non-trivial attack success rate remained at 0.0% even with normalized perturbations as high as 0.50. This suggests that single-step manipulations are effectively filtered out by the inertia of the strategy's calculation window.

Limitations

Some timing policies assume different attacker knowledge. For example, `high_volatility` requires only public market data, while `high_pred_volatility` and `near_strategy_boundary` require stronger knowledge of the ATS.